

Design note: sequential fertility with a priced biological clock

A nested modification of the intergenerational housing–fertility model

Fertility–housing project

July 18, 2026 — design for the `intergen_seq_fertility` fork; no production change

1. Purpose and scope

In the production model, fertility is a *one-shot family-size choice with a recurring clock*: at each fertile age a childless household’s κ_f -logit picks among remaining childless and forming a family of size $1, \dots, \bar{n}$; a chosen family arrives with certainty, lands in the just-gave-birth stage, and the fertility choice never fires again. The choice *recurs* period by period only while childless. What the model lacks is a *price of time*: family formation succeeds with certainty whenever chosen, so waiting is costless, the option value of postponement is degenerate, and every response of fertility to risk, prices, or policy loads on completed size rather than on timing. This note specifies the minimal modification that prices time — an age-declining conception probability — together with the new timing moments it makes measurable, the identification bookkeeping, and the verification protocol. The goal of the fork is a solver that is (i) bit-identical to production when the mechanism is off, (ii) tested, and (iii) ready for calibration and for a timing-versus-completed-fertility comparison against the production model. The payoffs we are buying, beyond fit: policy experiments gain a tempo/quantum decomposition (do birth grants create births or move them?), housing affordability gains a postponement-induced childlessness channel, and age-targeted policies (grants by age, entry-debt relief) acquire an economic rather than mechanical age profile. Steady-state comparisons reveal tempo in hazard shapes; full announcement-dynamics of tempo require transition paths, which remain out of scope here.

Three margins are deliberately *not* added in v1, each one line of why: contraception failure (unwanted births) would require family-planning moments we do not target; attempt costs (medical, search) are unidentified without ART data; within-couple bargaining (Doepke–Kindermann) changes the decision unit, which is a different paper.

2. The modification, as primitives

Current primitive. During fertile ages $j \in [A_f^{start}, A_f^{end}]$ a childless household at $(n, s) = (0, 0)$ faces a κ_f -smoothed logit over $\{\text{remain childless}\} \cup \{\text{form a family of size } n' \in \{1, \dots, \bar{n}\}\}$; the chosen family arrives with certainty at $(n', 1)$, child stages then age by the existing stage-transition machinery, and households with $n \geq 1$ never face the fertility choice again.

Modified primitive. The childless household chooses to *attempt* family formation at size n' . An attempt succeeds within the period with probability π_j , landing at $(n', 1)$ exactly as today; with probability $1 - \pi_j$ it fails, the household remains at $(0, 0)$, and may attempt again next period inside the window — retrieval is what the existing recurring choice already supports for free. The attempt

itself is costless in v1. The schedule π_j is a primitive of the environment,

$$\pi_j = \max\{0, \min\{1, 1 - \omega_1 e^{\omega_2 (a_j - A_f^{start})}\}\}, \quad \pi_j = 0 \text{ for } a_j \geq 45,$$

the four-year-period analogue of the exponential infertility risk in Sommer (2016), with (ω_1, ω_2) calibrated *externally* to demographic conception-within-window schedules by maternal age (Léridon-type fecundity tables aggregated to the model’s four-year periods), not estimated in the SMM. Setting $\pi_j \equiv 1$ recovers the production model identically; this nesting is the fork’s acceptance test.

Choice and value. Each family-formation alternative in the fertile-age logit replaces its certain continuation with the lottery

$$V_j^{attempt\ n'}(b, z, \cdot) = \pi_j V_j(b, z, n', 1, \cdot) + (1 - \pi_j) V_j(b, z, 0, 0, \cdot),$$

inside the same κ_f -logit over alternatives (the remain-childless alternative is unchanged). One sentence of economics: because π_j falls with age, the value of postponing an attempt now carries a real cost — the risk that the window closes — so delay is an insurance margin with a price, and households near the end of the window attempt more urgently than the same households earlier in life. Nothing else in the household problem changes: budget sets, housing menus, tenure choice, location choice, bequests, and the transfer floor added on July 18 are untouched.

Distribution. Wherever the population flow currently moves attempting mass into the birth state with certainty, it now splits it $\pi_j : (1 - \pi_j)$ between the birth destination and the unchanged state. The accounting identity “births into $(n + 1)$ at age $j = \pi_j \times$ attempting mass at j ” must hold exactly and is a unit test. Measured births everywhere in the model (birth grants, entrant creation from matured children, completed fertility entering bequest utility) follow realized births, not attempts, so all downstream objects update through the same realized flow with no further modification.

3. New measurement

All of the following are read off population flows; none require simulation. Timing resolution is one model period (four years) — stated once, honestly: age at first birth is measured in four-year bins, and spacing in whole periods.

- **Family-formation hazard** h_j : realized family formations at age j divided by childless mass at j ; the attempted hazard h_j^A (attempts over childless mass) is also computable, and $h_j = \pi_j h_j^A$ by construction. Because the production choice is one-shot in size, v1 has no second-birth hazard and no spacing distribution — families of size two form in a single event. Pricing *per-birth* sequencing (a real second-birth hazard) requires redesigning the child state to track the youngest child’s stage jointly with a running birth count; that is the costed v2 extension of Section 5, not part of this fork.
- **Age at first birth**: the distribution of realized family-formation flows across age bins; headline moments are the mean and the share of first births at ages 30 and above.
- **Childlessness decomposition**: completed childlessness splits into *chosen* (never attempted during the window) and *clock* (attempted at least once, never succeeded). Failed attempters and never-attempters occupy the same state, so the exact split requires carrying a shadow never-attempted mass through the population flow. In v1 the split is instead computed by integrating the measured attempt and realized hazards over a synthetic cohort — exact under

within-age homogeneity of attempt rates and labeled as an approximation otherwise; the exact shadow-mass accounting is a v1.1 refinement, and either way it is measurement, not a state: the household problem is unchanged.

- **Tempo/quantum readout for policy:** for any policy counterfactual, report the change in completed fertility (quantum) and the change in the birth-hazard age profile and mean age at birth (tempo). In steady-state comparisons the two are linked by cohort accounting; the decomposition becomes fully dynamic only on transition paths (out of scope).

4. Identification bookkeeping

New parameters: (ω_1, ω_2) and the window close — all *external* restrictions from demographic schedules; the SMM gains no free parameters. New usable moments: the age profile of first-birth hazards, mean age at first birth, the over-30 first-birth share, and the chosen/clock childlessness split at 45 — from NCHS/NSFG vital statistics and the PSID childbirth history (targets and standard errors to be constructed in a separate data pass; that pass is the remaining precondition for calibration). These moments over-identify the existing fertility block $(\psi, \kappa_f, \bar{c}_n, \bar{h}_n, \bar{h}_{jump})$, whose current discipline is levels and gradients only. One substitution risk to watch: the logit scale κ_f and the fecundity schedule both smooth hazards; they are separated because π_j is pinned externally by biology, leaving κ_f to the choice margin (income and price gradients of hazards). A known fixed- θ tension to expect: the clock adds a force for *early* births (attempt while π is high), while the production calibration generates late first births through savings and housing constraints; how these interact under recalibration is exactly the timing-versus-completed-fertility question the fork exists to answer.

5. Verification protocol and deliverables

- (i) *Nesting:* with $\pi_j \equiv 1$ the fork must reproduce the production M5 local baseline bit for bit (the same gate0 standard used for the July-18 transfer floor: loss and market residual to the last digit).
- (ii) *Unit tests:* fecundity-schedule arithmetic; the birth-flow accounting identity; hazard identities $h_j = \pi_j h_j^A$; conservation of mass across the $\pi : (1 - \pi)$ split; a revive/kill pair mirroring the floor-test pattern.
- (iii) *Smoke:* one GE solve at the M5 winner with a demographic π schedule, reporting the standard packet plus the new timing moments — the first quantitative look at what the clock does at fixed parameters.
- (iv) *Ready for calibration:* the fork solves, the timing moments exist as model output with a documented regeneration command, and the comparison protocol is: calibrate the fork on the production target system plus timing moments, then compare both models' policy experiments with the tempo/quantum readout of Section 3.

The fork lives in `code/model/intergen_seq_fertility/` (a renamed copy of the production package with its own tests and drivers); production code is not modified. Design decisions above that later prove wrong are cheap to revise — the fork is disposable by construction.

The v2 extension, costed. True per-birth sequencing — separate attempt decisions for the first and second child, a second-birth hazard, and a spacing distribution — requires the child state to track (birth count, stage of youngest child) rather than the current (family size, one stage clock), touching the transition tensor, the value-function shape, both Bellman implementations, three population-flow implementations, and every (n, s) -keyed policy hook. It is the natural next step if the v1 comparison shows timing margins matter, and a separate scope decision: v1's attempt

machinery, fecundity schedule, timing moments, and tests all carry over to it unchanged.